# Human Activity Recognition from Wi-Fi CSI Doppler Spectrograms: A CNN Baseline and a Contrastive Extension for Cross-Domain Generalization

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***Abstract*—We study device-free human activity recognition (HAR) from the Doppler spectrograms of Wi-Fi channel-state information (CSI), using the publicly released traces of the SHARP dataset [1], [2]. Starting from the Inception-style convolutional baseline of the course material—trained *only* on measurement set $S1$ and evaluated on the six unseen sets $S2$–$S7$, where it reaches $48.2\%$—we add and measure, one at a time, three strategies for closing the cross-domain gap without the hand-engineered front-end of the reference system. (i) *Supervised contrastive pre-training*: a batch-normalized encoder trained with a SupCon objective and domain-randomization augmentations, with a linear probe on the frozen representation, gives $53.1\%$ and improves *every* measurement set, a t-SNE analysis confirming that the learned space is environment-invariant. (ii) *A slower optimizer schedule* reaches $51.3\%$ while fitting the training environment *worse*, showing that on a single-environment training set the two objectives conflict. (iii) *Multi-antenna soft decision fusion*: a recording-level group identifier threaded through the data pipeline lets the four per-antenna probability vectors of one physical time window be averaged, lifting accuracy to $61.5\%$ ($+10.2$ points, $+7.4$ to $+17.4$ per set) with no retraining and no extra parameters—the largest gain we measured, enough to overtake the single-component ablation of the reference system. We also report the front-end work that makes an overlapping ($91\%$) sliding window affordable in memory. The residual gap to the full pipeline ($> 95\%$) is attributable to its strongest-path phase sanitization, which we do not implement.**

***Index Terms*—Human Activity Recognition, Wi-Fi Sensing, Channel State Information, Convolutional Neural Networks, Contrastive Learning, Domain Generalization.**

## I. Introduction

Contactless sensing of human activity is valuable in healthcare, security and ambient monitoring. Commodity Wi-Fi devices can be repurposed as radar-like sensors: as a person moves, their body induces micro-Doppler shifts captured in the channel-state information (CSI) estimated by the receiver, *without recording any video*. After processing, the CSI becomes a *Doppler spectrogram*—a time $\times$ velocity image—whose appearance changes markedly with the activity, so recognition can be cast as image classification with a convolutional neural network (CNN) [1].

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Course project for Neural Networks and Deep Learning. Baseline pipeline by M. Carraro; Supervised Contrastive Learning extension by F. Vezzani.

The central difficulty is *generalization*. Different people move differently, and different rooms (and even the same room on different days) distort the signal differently, so a model that classifies perfectly in its training environment may collapse on a new one. This is the challenge raised in the project proposal, which explicitly suggests a *contrastive / self-supervised* approach to obtain subject- and environment-independent features. Prior systems close this gap mostly through signal processing—robust phase sanitization and multi-antenna fusion—at the cost of a heavy, hand-engineered pipeline [1]; we instead ask how far a purely *representation-learning* remedy can go on top of a minimal front-end.

In this report we tackle five-class Wi-Fi HAR under a strict cross-domain protocol (train on a single environment, test on six unseen ones) and ask how much of the domain gap can be closed *without* the hand-engineered front-end: first by *learning* invariance with Supervised Contrastive Learning (SupCon) [3] instead of engineering it, then by re-using the receiver diversity that the dataset already provides. Both remedies are data-efficient—they reuse the same small training set and a lightweight encoder—and both compose with the orthogonal signal-processing tools of the reference system. Every change is implemented and measured *one at a time*, on top of the same baseline and the same protocol, so that each accuracy figure is attributable to one design decision (Tab. 1). The contributions are:

- **Baseline (M. Carraro).** A faithful reimplementation, in plain PyTorch, of the dataset pipeline and of the baseline Inception-style CNN, with a cross-domain evaluation protocol (train on $S1$, test on $S2$–$S7$) and a per-measurement-set error breakdown.
- **Contrastive representation (F. Vezzani).** A deeper batch-normalized encoder pre-trained with a supervised-contrastive loss and domain-randomization augmentations, then evaluated with a linear probe on the frozen features. It improves cross-domain accuracy on every set.
- **Multi-antenna decision fusion.** A group identifier that survives the whole data pipeline recovers which windows are the *same* physical time window seen by the four receiver antennas, so their predictions can be fused by averaging class probabilities—the largest gain we mea-

sured, with no retraining.

- **Front-end and optimization engineering.** A sliding window with the stride used by the SHARP authors (30 samples, 91% overlap) replaces the naïve non-overlapping segmentation, supported by a two-pass, memory-bounded dataset builder; and a controlled comparison of learning-rate and early-stopping settings shows that, with a single training environment, fitting the training set better does *not* imply generalizing better.

We use the full SHARP system [1] as the reference upper bound and discuss, quantitatively, which part of the gap each strategy removes and which part is left.

Section II reviews the related work; Section III gives a high-level view of the pipeline and of the changes we evaluate; Section IV details signals and features and Section V the learning frameworks. Results and the comparison against SHARP are in Section VI, concluding remarks in Section VII.

## II. Related Work

**Wi-Fi sensing and the SHARP system.** Device-free HAR from commodity Wi-Fi has matured rapidly thanks to CSI extraction tools and public datasets. The most relevant work for us is SHARP [1], which recognizes activities from the Doppler information in the CSI of IEEE 802.11ac links and targets *environment and person independence* through two ingredients: (i) a robust phase-sanitization step that isolates the multipath components and adopts the *strongest path* as phase reference, and (ii) a *decision-fusion* scheme that runs a compact Inception-style CNN on each of the four receiver antennas and combines their decisions. On the seven measurement sets of the accompanying dataset [2] the full system exceeds 95% even on rooms, days and persons never seen at training time. Those results rest on a heavy, hand-engineered front-end: the paper's own ablation shows that replacing the strongest-path sanitization with the standard reference-antenna method makes the *same* CNN collapse to 39–60% on the unseen sets, i.e. much of the robustness is carried by the pre-processing rather than by the learned representation. Our work keeps the minimal front-end (no strongest-path sanitization) and asks how much cross-domain robustness can be recovered elsewhere: on the *representation* side (contrastive pre-training) and on the *decision* side, by re-implementing SHARP's antenna fusion—which, unlike the sanitization step, costs nothing at training time and turns out to be worth more than any other change we made.

**Contrastive representation learning.** Contrastive methods learn representations by pulling together "views" that should be similar and pushing apart those that should differ. SimCLR [4] established the self-supervised recipe: two augmentations of an image form a positive pair, all other images in the batch are negatives, and an encoder is trained with a temperature-scaled InfoNCE loss on $L_2$-normalized embeddings. Supervised Contrastive Learning (SupCon) [3] extends it to the labelled setting—*all* samples sharing the anchor's label become positives—producing tighter, more class-consistent clusters than cross-entropy. We adopt SupCon because its notion of "positive" maps onto our goal: two windows of the same activity should be neighbours *regardless of the environment, person or antenna they come from*. Combining SupCon with explicit domain-randomization augmentations for cross-domain Wi-Fi HAR on the SHARP data is the extension we contribute: it complements the signal-processing route of [1] by attacking domain shift from the feature-learning side, and we quantify its gains and its residual limits in Section VI.

## III. Processing Pipeline

All of our models cast Wi-Fi HAR as the classification of Doppler-spectrogram images and share the *same* data front-end; they differ in *how* the representation is learned and in *how* the per-window decisions are combined (Fig. 1). The raw CSI Doppler traces are cut into fixed-size windows of shape $1 \times 340 \times 100$, one antenna at a time (Section IV); from there the pipeline forks:

- the **baseline chain** (M. Carraro) feeds each window to an Inception-style CNN trained end-to-end with cross-entropy, and reads the activity off the softmax output;
- the **contrastive chain** (F. Vezzani) first applies domain-randomization augmentations to build two views of each window, pre-trains a deeper batch-normalized encoder with a Supervised Contrastive loss, then *freezes* the encoder and fits a linear probe on the resulting features;
- the **fusion stage** sits *after* either classifier and needs no retraining: the per-antenna class probabilities of the same physical time window are averaged into one decision.

All chains are trained on a single environment ($S1$) and evaluated under the same cross-domain protocol (unseen $S2$–$S7$), so they can be compared on equal footing against the reference SHARP system [1]; Section V details them.

### A. The changes we evaluate

Four modifications were implemented on that skeleton, each isolated so that its effect is attributable (Tab. 1): a different *windowing* strategy, a *group-aware* pipeline enabling *decision fusion*, a *contrastive* representation, and a different *optimizer schedule*. Only the first touches the data loader and only the third trains a new network; fusion is post-processing on an unchanged model, which is what makes its measured gain remarkable.

## IV. Signals and Features

*(Dataset pipeline implemented by M. Carraro.)*

### A. The SHARP data

We use the pre-extracted Doppler traces released with the SHARP dataset [2] (IEEE 802.11ac, 80 MHz). Each trace is an array of shape $(T, 100)$—$T$ time steps, 100 Doppler-velocity bins—from a single receiving antenna (`stream_1...stream_4`) of a four-antenna monitor. The recordings map to seven *measurement sets* $S1$–$S7$, each a

TABLE 1: The modifications evaluated, the code each required, and the resulting cross-domain accuracy (train on $S1$, test on $S2$–$S7$, one run). Each is measured against the reference configuration of the first row.

| Strategy | Implementation | Acc. |
|---|---|---|
| *Reference* | non-overlapping windows (stride 340, 13,564 windows); reduced-Inception CNN (128,535 par.); Adam $10^{-3}$, patience 7 | 48.2% |
| Contrastive | augmentations + BN encoder with global average pooling + SupCon loss; linear probe on the frozen encoder | 53.1% |
| Slow schedule | Adam $10^{-4}$, patience 5, budget 100; nothing else changes | 51.3% |
| Antenna fusion | group id emitted by the loader and carried through the split; softmax and per-group probability average at inference, same weights | **61.5**% |
| Sliding window | stride 30 (91% overlap, $\approx 10\times$ windows); two-pass pre-allocating loader, splits built separately | pending |

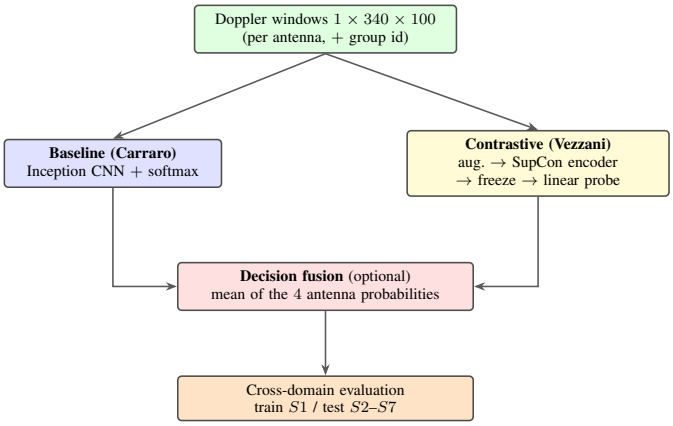



Fig. 1: Processing pipeline. A shared front-end turns CSI traces into Doppler windows tagged with a group id; the baseline chain classifies them directly, the contrastive chain learns an environment-invariant representation before a linear probe, and fusion collapses the four per-antenna decisions of a time window into one.

distinct (environment, monitor position, person, day) combination. Only $S1$ is used for training; the others probe generalization to an unseen day ($S2$), an unseen person ($S3$), blocked line-of-sight ($S4$, $S5$), a different room ($S6$), and the worst case of new room, day and person at once ($S7$).

### B. From traces to windows

To keep the setup comparable across *all* environments we use the five core activities present everywhere: **E** (empty room), **J** (jumping), **L** (sitting still), **R** (running), **W** (walking), matching the five-class task of [1]. Each trace is cut into windows of $W_{\text{len}} = 340$ time samples ($\approx 2\,\text{s}$) advancing by a stride $S$, plus a channel dimension: inputs of shape $1\times340\times100$ (Fig. 4a). Each antenna stream is an independent labelled sample, as SHARP does at training time; antenna identity is re-used only at *decision* time (Section V-C).

**Windowing strategy.** The reference configuration uses $S = W_{\text{len}} = 340$, i.e. *non-overlapping* windows: simplest, but wasteful—a trace of length $T$ yields only $\lfloor T/340 \rfloor$ examples and any pattern straddling a boundary is cut in half. It gives 13,564 windows, reasonably balanced (E: 3012, J: 2160, L: 3056, R: 2684, W: 2652). The alternative is a sliding window: after prototyping $S = 170$ (50% overlap) we adopted $S = 30$, the value hard-coded in the SHARP authors' own dataset script, i.e. 91% overlap and $\approx 10\times$ as many windows from the same recordings. Overlapping windows are strongly correlated, so the number of *independent* examples grows far less than tenfold, and a window and its neighbour can land on opposite sides of the random train/val split, making the validation loss optimistic and perturbing *when* early stopping fires. The test figures are unaffected: the test set is a disjoint set of environments.

**Group identifiers.** The four streams of a recording have identical length, so window $k$ of each starts at the same sample $kS$ of the same event. To exploit this, the parser keeps the full recording token (`"J2"`, repetition number included) and the loader emits a group id `env_token_k` per window: windows sharing it are the same physical time window seen by different antennas, which is what makes Section V-C possible. The id is carried through the split alongside the labels, and the test `DataLoader` uses `shuffle=False` to stay aligned with it.

**A memory-bounded loader.** At $S = 30$ the windows no longer fit comfortably in a Colab runtime: the original loader accumulated per-file arrays in a list and called `np.concatenate`, holding list *and* result in RAM at once—affordable at 1.8 GB, not at $\approx 20$ GB. The rewritten loader makes two passes: the first counts the windows each trace will produce, the second fills one pre-allocated `float32` array and frees each trace as soon as it is windowed. It also takes the environments to load, so train/val and test arrays come from two independent calls instead of one large array being sliced, and the copy left by `train_test_split` is deleted explicitly. Peak memory is roughly halved and the windows are identical.

### C. Cross-domain split

Following the SHARP protocol, the split is by environment, never random: the folders of $S1$ ($S1a, S1b, S1c$) are used for training/validation and every other set is held out for testing; a random split would leak windows from the same person/room/day into both train and test and overestimate performance. Within $S1$ we keep 85% for training and 15% for validation (stratified), giving 2767 / 489 / 10,308 windows for train / val / test at $S = 340$; the 10,308 test windows collapse to 2577 distinct physical time windows once the four antennas are grouped, which is the number of decisions the fused evaluation reports.

## V. Learning Framework

### A. Baseline CNN (M. Carraro)

The baseline reproduces the simplified reduced-Inception module of SHARP ( [1], Fig. 4), shown in Fig. 2: three parallel

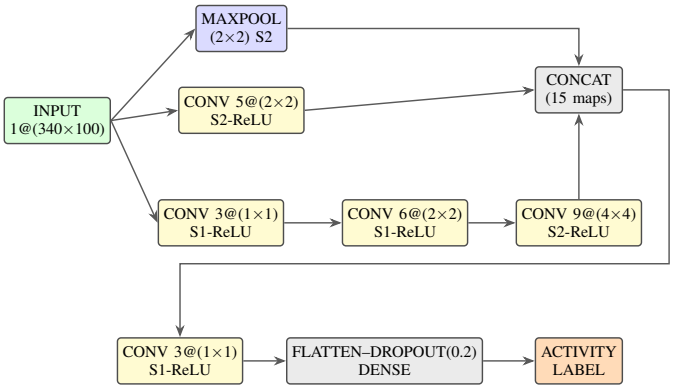



Fig. 2: Baseline single-antenna architecture, a faithful reproduction of the SHARP reduced-Inception module ( [1], Fig. 4). S2 denotes stride 2; every CONV is followed by a ReLU (128,535 parameters).

TABLE 2: The optimizer schedules compared; everything else (architecture, data, seed, batch size, split) is held fixed. "Stop" is the epoch early stopping fired at, in parentheses the epoch whose weights were restored.

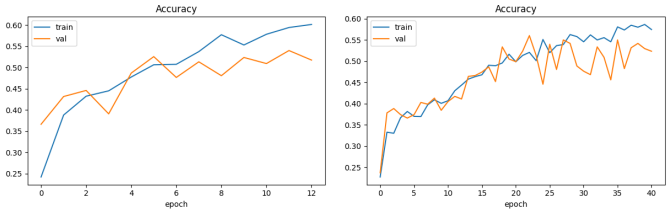
| Schedule | lr | patience | budget | stop (best) | train acc. |
|---|---|---|---|---|---|
| fast | $10^{-3}$ | 7 | 40 | 13 (6) | 0.601 |
| slow | $10^{-4}$ | 5 | 100 | 41 (36) | 0.575 |
| adopted | $10^{-3}$ | 12 | 100 | — | — |

branches (max-pool, strided conv, three stacked convs) turn the $1 \times 340 \times 100$ input into feature maps of identical size $170 \times 50$, concatenated on the channel axis ($1{+}5{+}9 = 15$), reduced to 3 channels by a $1 \times 1$ convolution, flattened, and classified by a dense layer after $\mathrm{Dropout}(0.2)$. The model has **128,535** trainable parameters—exactly the count reported for the SHARP baseline CNN [1]. Training uses cross-entropy, Adam, mini-batches of 64 and *early stopping*, restoring the lowest-validation-loss weights before testing.

**Two optimizer schedules** were compared with everything else held fixed (Tab. 2), because their effect is counter-intuitive. The *fast* schedule (lr $= 10^{-3}$, patience 7) stops at epoch 13 (best validation loss 0.993) with the classic overfitting signature of Fig. 3(a): training accuracy climbs to 0.60 while validation accuracy plateaus near 0.52. The *slow* schedule (lr $= 10^{-4}$, patience 5, budget 100) runs for 41 epochs with the two curves almost on top of each other (Fig. 3(b)) and stops at a training accuracy of only $\approx 0.58$: the model has not yet fitted its own training set. Reading that as underfitting, the final configuration keeps lr $= 10^{-3}$ but raises patience to 12, so a plateau no longer ends training prematurely—which matters at stride 30, where each epoch is an order of magnitude slower.

### *B. Contrastive extension: Supervised Contrastive Learning (F. Vezzani)*

*1) Motivation:* The baseline is trained with plain cross-entropy on a single environment, so nothing pushes its features to be invariant across domains. Following the *contrastive / self-supervised* direction suggested in the project proposal, we implement *Supervised Contrastive Learning* (SupCon) [3]. The key change is *what* is optimized: instead of predicting the label directly, an *encoder* is trained so that two windows of the *same activity* are pulled together and windows of different activities are pushed apart—*regardless of the environment, person or antenna they come from*. Because the objective depends only on activity identity, the encoder is driven to discard environment-specific nuisances; a tiny linear classifier is then fit on the frozen representation (*linear probe*).



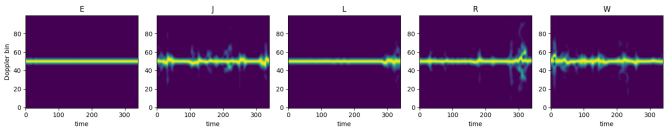

(a) fast: $10^{-3}$, pat. 7 (b) slow: $10^{-4}$, pat. 5

Fig. 3: Baseline train/validation accuracy under the two optimizer schedules. In (a) the training curve detaches from the validation one after $\approx 6$ epochs (overfitting the single environment); in (b) the two stay glued together and stop at a training accuracy of 0.575—*under*-fitted, and yet generalizing better (Tab. 3).

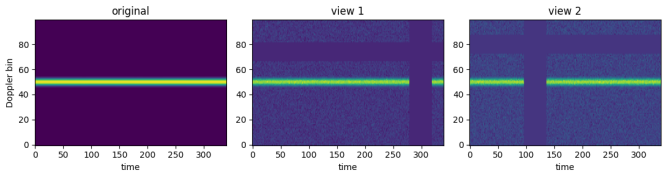
*2) Domain-randomization augmentations:* SupCon needs, for each window, two augmented "views" that keep the activity intact but perturb nuisance factors, acting as a light domain randomization (Fig. 4b). All transforms run on-GPU on a whole batch (default parameters in parentheses):

- **amplitude scaling** by a per-window random gain ($\mathcal{U}(0.8, 1.2)$)—rooms/distances reflect the signal with different strength;
- **additive Gaussian noise** ($\sigma = 0.05$);
- **circular time shift** ($\pm 40$ samples)—the activity does not depend on where in the $\approx 2$ s window it starts;
- **time and Doppler masking** (SpecAugment-style: a zeroed band of 40 time samples and 15 velocity bins)—forces the encoder to use the whole time–velocity pattern rather than any single region.

*3) Encoder and projection head:* The encoder $f(\cdot)$ is a deeper, batch-normalized evolution of the baseline's Inception block. It keeps the three-branch front-end (max-pool / strided conv / deeper conv, concatenated to 41 channels) and adds (i) batch normalization after every convolution, (ii) a $1 \times 1$ channel-mixing reduction to 32 channels, (iii) a second down-sampling stage ($32 \to 64 \to 128$, output $128 \times 43 \times 13$), and (iv) *global average pooling*, which replaces the baseline's large `flatten`→dense layer with a fixed 128-dimensional representation $h$ that has fewer parameters and is more shift-robust. A small MLP *projection head* $g(\cdot)$ ($128 \to 128 \to 128$, ReLU) maps $h$ to an $L_2$-normalized embedding $\boldsymbol{z} = g(h)$ on the unit hypersphere; as prescribed by SupCon, the loss is applied to $\boldsymbol{z}$ and the head is *discarded* after pre-training, so classification uses $h$. The whole model has 130,952 parameters, comparable




(a) one input window per activity




(b) a window and two augmented views

Fig. 4: (a) One Doppler-spectrogram window per activity (transposed: rows are the 100 velocity bins, columns the 340 time samples); low-motion classes concentrate energy at zero velocity, *running/jumping* spread it widely. (b) Scaling and noise change the texture and the circular shift and SpecAugment-style masks (dark bands) hide parts of the pattern, while the activity signature survives.

to the baseline's 128.5k.

*4) SupCon loss:* For a batch of $L_2$-normalized embeddings with labels, each anchor $i$ attracts every other sample sharing its label (its positive set $P(i)$) and repels all others, through a temperature-scaled softmax over cosine similarities:

$$\mathcal{L} = \frac{1}{|I|} \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\boldsymbol{z}_i \cdot \boldsymbol{z}_p / \tau)}{\sum_{a \neq i} \exp(\boldsymbol{z}_i \cdot \boldsymbol{z}_a / \tau)}, \qquad (1)$$

where $I$ is the set of anchors with at least one same-class positive (the implementation averages the per-anchor terms over the batch) and the temperature is $\tau = 0.07$. Unlike self-supervised SimCLR [4], using the labels in Eq. (1) makes *all* same-activity windows in the batch positives—even those from different files or environments—which turns generic contrastive learning into an environment-invariant *activity* representation.

*5) Training and evaluation protocol:* Everything uses the *same* $S1$-based split as the baseline. In **pre-training**, each batch is expanded into two augmented views and encoder and head are trained jointly with Adam (lr $10^{-3}$, weight decay $10^{-4}$) for up to 30 epochs with early stopping (patience 7) on the validation contrastive loss. The **linear probe** then freezes the encoder, pre-computes and standardizes the embeddings, and trains a single $\mathrm{Linear}(128, 5)$ layer with cross-entropy—isolating the quality of the representation—while **optional fine-tuning** unfreezes the encoder and trains end-to-end at lr $10^{-4}$. A t-SNE projection of the test embeddings, colored by activity and by environment, serves as the invariance diagnostic. A variant that lowered the contrastive and probe rates to $10^{-4}$ alongside the baseline's was interrupted during pre-training and is not reported.

### C. Multi-antenna decision fusion

The dataset ships the four antenna streams *separately*, and both chains so far discard that structure. SHARP keeps the per-antenna training convention but fuses the four decisions at inference [1]; that step is reproduced here as pure post-processing—classifier, weights and training procedure untouched. For a group id $g$, let $A_g$ be the (up to four) antenna windows carrying it and $\boldsymbol{p}_i = \mathrm{softmax}(f(\boldsymbol{x}_i))$ the class probabilities of window $i$. The fused decision is

$$\hat{y}_g = \underset{c \in \{1,\ldots,5\}}{\arg\max} \; \frac{1}{|A_g|} \sum_{i \in A_g} p_{i,c}, \qquad (2)$$

i.e. a *soft* vote, preferred to majority voting on the hard labels because a confident antenna can outweigh two unsure ones and because it degrades gracefully when a recording provides fewer than four streams. The reference paper does not detail its own fusion rule, so Eq. (2) is a standard instantiation of the idea rather than a byte-for-byte reproduction.

Two details make it correct: the group id must *not* encode the stream index, so the four antennas collide onto one key; and the evaluation `DataLoader` must not shuffle, or the probabilities stop lining up with the stored ids and the average silently mixes unrelated windows. Fusion also changes the *unit* of evaluation: 2577 physical time windows instead of 10,308 antenna-level ones.

## VI. Results

All numbers come from the stored notebook outputs, one run per configuration, under the same protocol: train on $S1$, test on the unseen $S2$–$S7$. Tab. 3 collects them per measurement set and Fig. 7 plots them; Tab. 1 maps each column to the change that produced it.

### A. Baseline

On the pooled unseen test set the baseline reaches $\mathbf{48.2}\%$ accuracy (test loss 1.052), from $45.5\%$ ($S2$) to $52.3\%$ ($S7$). Its confusion matrix (Fig. 5 (a)) shows two dominant error modes: *sitting still* is absorbed by *empty* ($L \to E$: 0.76), both being near-static and concentrated at zero Doppler velocity, and *running/walking* are heavily confused with each other (0.39 and 0.28).

### B. Contrastive representation

The linear probe on the frozen SupCon encoder raises overall accuracy to $\mathbf{53.1}\%$ (validation $58.3\%$): $+4.9$ points, and an improvement on *every* set, the largest on $S3$ ($+8.3$, unseen person) and $S7$ ($+5.9$, all conditions new). Its confusion matrix (Fig. 5 (b)) sharpens *running* and softens the $L \to E$ confusion ($0.76 \to 0.52$).

The t-SNE projection (Fig. 6) is the most informative diagnostic. Colored by *environment* (right) the points are thoroughly mixed—the SupCon objective *did* reach its design goal of an environment-invariant representation. Colored by *activity* (left) the space splits into a low-motion and a high-motion lobe, but the activities *within* each lobe stay partly entangled: contrastive learning removed the *domain* nuisance and exposed a harder, intrinsic limit—a single-antenna $\approx 2$ s window does not fully separate the activities, whichever way it is encoded. Consistently, the SupCon loss fell only from $\sim 4.6$

to $\sim 4.4$ with a noisy validation curve. *End-to-end fine-tuning* then *hurts*, dropping back to $48.4\%$: freezing the pre-trained encoder is what generalizes, while fine-tuning re-overfits the single training environment.

### C. Optimizer schedule

The slow schedule changes only the learning rate, the patience and the epoch budget (Tab. 2), so the two runs isolate that factor. Slowing training down *improves* cross-domain accuracy from $48.2\%$ to $51.3\%$ ($+3.1$ points, five sets out of six) even though the model fits its own training data *worse* ($0.575$ vs. $0.601$, Fig. 3). With a single training environment the two objectives conflict: an epoch spent fitting $S1$ better is also an epoch spent memorizing its room, person and day. It also tempers the previous subsection—$+3.1$ of the contrastive model's $+4.9$ points can be bought with an optimizer setting alone, so single-run gains of a few points here are fragile.

### D. Multi-antenna decision fusion

Averaging the four per-antenna probability vectors (Eq. (2)) lifts the *same* model from $51.3\%$ to $\mathbf{61.5}\%$: $+10.2$ points overall, with per-set gains between $+7.4$ ($S6$) and $+17.4$ ($S7$) and no set left behind (Fig. 7), for one softmax and one group-average at inference—no retraining, no extra parameters.

The mechanism is visible in Fig. 5 (c–d): fusion does not repair a broken class, it sharpens decisions the single-antenna model already leaned towards (*empty* $0.82 \rightarrow 0.89$, *jumping* $0.50 \rightarrow 0.54$, *running* $0.46 \rightarrow 0.61$, *walking* $0.41 \rightarrow 0.57$). The antennas observe the same motion through different multipath, so their errors are only partially correlated and averaging cancels part of them. The exception is the low-motion pair: $L$ stays confused with $E$ ($0.54$ of its windows) because all four antennas agree on the same wrong answer—with almost no motion, extra spatial views add no information, and that residual error belongs to the $2$ s window representation, not to the decision rule.

### E. Sliding-window front-end

This is the one change we cannot yet quantify: the notebook adopting it as the default ships with its outputs cleared, and the $50\%$-overlap prototype was never executed end-to-end. The code does establish the mechanism and its cost—the same recordings yield $\approx 10\times$ more windows, at $\approx 20$ GB of resident memory and an epoch time an order of magnitude longer, exactly why the loader had to allocate once instead of concatenating. Whether those correlated extra windows convert into accuracy is left open: SHARP's own use of stride $30$ suggests they help, but we report the change as implemented, not as measured.

### F. Comparison with the SHARP paper

Tab. 3 places our results next to the SHARP paper [1]: the *full* system (strongest-path sanitization + 4-antenna fusion) and an *ablation* that swaps the strongest-path sanitization for the standard reference-antenna method, feeding the *same* CNN ( [1], Table 4). Full SHARP stays above $95\%$ everywhere. Our contrastive model ($53.1\%$) reaches the level of that ablation ($\sim 53\%$ mean); the fused model ($61.5\%$) overtakes it and, on the blocked-line-of-sight sets $S4$ and $S5$ where the ablation collapses to $43.5\%$ and $39.2\%$, beats it by $14$ to $18$ points. The residual $\sim 35$-point gap is therefore concentrated in the one ingredient we do not implement—the strongest-path phase sanitization—and no longer in the fusion step. The acquisition campaigns behind the dataset also spanned nearly two years ( [1], Sec. 5.2), and that time diversity is itself part of what the reference results build on.

TABLE 3: Per-set accuracy (%), training on $S1$: single-antenna baseline under the two schedules of Tab. 2, SupCon linear probe, and the slow-schedule baseline with 4-antenna fusion (same weights). SHARP columns from [1] (Tables 3 and 4).

| | **Ours (this work)** | | | | **SHARP [1]** | |
|---|---|---|---|---|---|---|
| Set | Base (fast) | SupCon (probe) | Base (slow) | Fused | Full (fusion) | Abl. (1 ant.) |
| $S2$ | 45.5 | 51.1 | 50.9 | 62.6 | 99.8 | 59.8 |
| $S3$ | 51.4 | 59.7 | 55.9 | 68.8 | 99.3 | 58.0 |
| $S4$ | 48.5 | 51.9 | 50.6 | 58.1 | 97.6 | 43.5 |
| $S5$ | 47.5 | 52.9 | 49.8 | 57.5 | 96.6 | 39.2 |
| $S6$ | 48.1 | 51.3 | 50.6 | 57.9 | 99.8 | 59.4 |
| $S7$ | 52.3 | 58.2 | 51.5 | 68.9 | 96.0 | 60.3 |
| **Overall** | **48.2** | **53.1** | **51.3** | **61.5** | $\sim 98$ | $\sim 53$ |

### G. Discussion

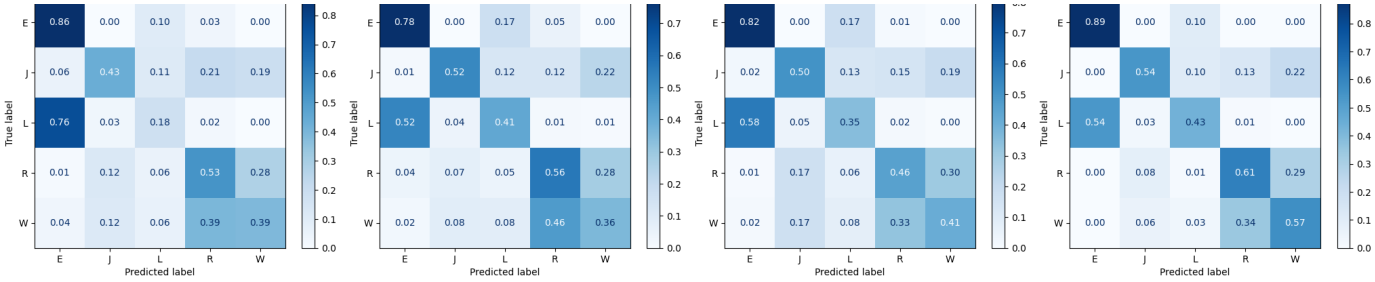
*a) What each change bought:* Ranked by measured effect, decision fusion ($+10.2$ points) dominates the contrastive representation ($+4.9$) and the optimizer schedule ($+3.1$). The ordering is itself a finding: the cheapest change—a few lines of post-processing over information already in the dataset—beat the one that cost the most design effort, consistently with SHARP's ablation, where accuracy grows monotonically with the number of fused antennas [1].

*b) Why we are still far from SHARP:* The gap is now essentially one component. SHARP uses the *strongest* multipath component as phase reference, and its ablation shows the standard reference-antenna method makes the *same* CNN collapse from $> 95\%$ to 39–$60\%$ (Tab. 3, last column)—the band our models occupy, ours at its top: we have reproduced the decision-level half of SHARP's robustness and none of its signal-processing half. Training on one small environment also makes every number high-variance.

*c) The two remedies are complementary:* The SupCon stage attacks domain shift on the *representation* axis and, as the t-SNE shows, succeeds there; its residual errors are *activity* confusions, exactly what fusion attacks. The experiment left undone is Eq. (2) on top of the frozen probe with the sliding-window front-end: all three components exist in the released code, but no run combines them.

## VII. Concluding Remarks

Starting from a baseline Wi-Fi HAR pipeline evaluated under a strict cross-domain protocol ($48.2\%$ on unseen environments), we implemented and measured three strategies one at a




(a) baseline, fast schedule (b) SupCon linear probe (c) baseline, slow schedule (d) + 4-antenna fusion

Fig. 5: Row-normalized confusion matrices on all unseen test environments, one per configuration. The contrastive representation (b) softens the $L \to E$ confusion of the baseline (a); fusion (d) raises every motion class of (c) with the same weights, but leaves *sitting*→*empty* in place.

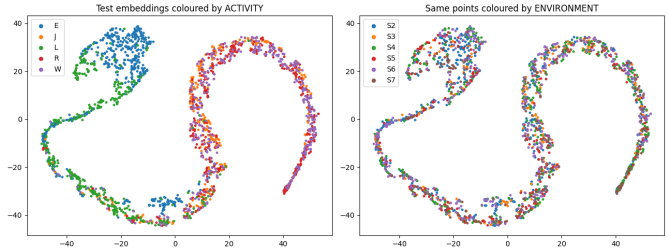



Fig. 6: t-SNE of the test embeddings of the SupCon encoder. *Left*, colored by activity: a low-motion and a high-motion lobe. *Right*, the same points colored by environment: the colors are mixed *inside* the structure, i.e. the space is organized by *activity*, not by *environment*.

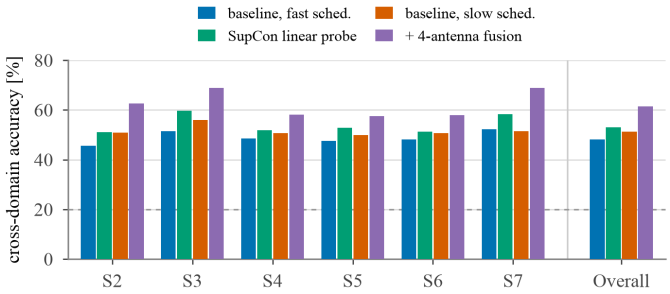



Fig. 7: Cross-domain accuracy per measurement set for the four configurations of Tab. 3 (dashed line: the 20% chance level). Every unseen set benefits from the contrastive representation and, much more, from antenna fusion.

time: a supervised-contrastive representation (53.1%, better on every measurement set, with a t-SNE confirming environment-invariance), a slower optimizer schedule (51.3%), and multi-antenna soft decision fusion, which lifts the same weights to 61.5% and overtakes the single-component ablation of the reference system [1].

The two effective remedies act on different axes and cost almost nothing: the contrastive encoder recovers domain robustness on the learning side and yields a frozen, reusable feature space, while fusion recovers accuracy on the decision side from receiver diversity the dataset already contains. Combining them—fusion on top of the frozen probe, with the sliding-window front-end—is the immediate next step, together with multi-seed runs and the strongest-path sanitization that still separates us from the full reference pipeline.

*a) What we learned and difficulties encountered:* The strict environment-based split changes the story: modest-looking numbers ($\sim$ 50%) are the honest cost of never seeing the test environment, which a random split would have hidden. Two results were counter-intuitive—training *less* well generalized *better*, and fine-tuning the contrastive encoder *hurt* by re-overfitting the single environment, which taught us to trust the frozen-encoder protocol—and the largest gain came from the cheapest change, once the data pipeline carried a group identifier. The main difficulties were getting SupCon pre-training to converge (the loss barely moved), the run-to-run variance of a small model trained on one environment, and the memory cost of overlapping windows, which forced a pre-allocating data loader before the 91%-overlap front-end could run at all.

## Author Contributions

**Marco Carraro:** dataset/windowing pipeline, baseline Inception CNN, training with early stopping, cross-domain evaluation. **Francesco Vezzani:** the Supervised Contrastive Learning extension—augmentations, encoder and projection head, SupCon loss, pre-training, linear-probe / fine-tuning evaluation, t-SNE. The remaining changes of Tab. 1—sliding-window front-end, group-aware pipeline with decision fusion, optimizer study—were carried out jointly.

## References

[1] F. Meneghello, D. Garlisi, N. D. Fabbro, I. Tinnirello, and M. Rossi, "SHARP: Environment and Person Independent Activity Recognition With Commodity IEEE 802.11 Access Points," *IEEE Transactions on Mobile Computing*, vol. 22, pp. 6160–6175, Oct. 2023.

[2] F. Meneghello, N. D. Fabbro, D. Garlisi, I. Tinnirello, and M. Rossi, "A CSI Dataset for Wireless Human Sensing on 80 MHz Wi-Fi Channels," *IEEE Communications Magazine*, vol. 61, pp. 146–152, Sept. 2023.

[3] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan, "Supervised Contrastive Learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 18661–18673, Dec. 2020.

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